Week 17 — Extended Reading: Exploration and adaptation

STUDY-RL

Source audit: 20 October 2026

Read for a claim, an assumption and a test

Read the teaching guide first, then use this pack alongside the lecture. The annotations below are course-written paraphrases and investigation prompts, not quotations from the papers. Each source link identifies a primary paper or official guidance. Benchmark results do not transfer automatically to our small exercises. For each reading, write the mathematical object, the needed assumptions and one observation that would challenge your interpretation.

1 Burda et al. (2018): RND

Primary source. https://arxiv.org/abs/1810.12894

Mechanism and scope. RND rewards prediction error against a fixed randomly initialised target network. The predictor learns visited observations; the target must remain fixed. Novelty in representation space need not correspond to progress on the external task.

Read, derive, investigate. Check that target parameters receive no optimiser update. Revisit one observation repeatedly and track its prediction error. Then change the observation representation: why can new irrelevant pixels look novel? The exact normalisation convention is part of the algorithm, not a cosmetic unit change.

2 Pathak et al. (2017): ICM

Primary source. https://arxiv.org/abs/1705.05363

Mechanism and scope. ICM learns features through an inverse-dynamics objective and uses forward prediction error as intrinsic reward. The representation aims to emphasise controllable aspects, but that is a modelling choice rather than a universal immunity to irrelevant noise.

Read, derive, investigate. Compare predicting raw observations with predicting learned features. What information does the inverse action-prediction task preserve or discard? Design a noisy observation channel and identify which assumption would make it irrelevant to the learned representation.

3 Badia et al. (2020): Agent57

Primary source. https://arxiv.org/abs/2003.13350

Mechanism and scope. Agent57 combines multiple exploration and long-horizon learning components with adaptive choice among policies. Its Atari result evaluates the complete system, not an isolated proof that a particular intrinsic bonus always improves exploration.

Read, derive, investigate. Build a component map: replay, recurrent state, intrinsic rewards, discount choices and policy selection. Which components are absent from the workbook? State one ablation needed before attributing the full system's result to RND-like novelty alone.

4 Finn et al. (2017): MAML

Primary source. https://arxiv.org/abs/1703.03400

Mechanism and scope. MAML learns parameters that perform well after a small adaptation update on a task. The outer loss differentiates through the inner update in full MAML. First-order approximations omit part of that derivative.

Read, derive, investigate. For $L_m(\theta) = \frac{1}{2}(\theta - m)^2$, derive the adapted parameter and outer gradient. Explain the factor $(1 - \alpha)^2$. In a softmax bandit, adding a constant to every logit changes no probability, so symmetry does not establish a unique optimal parameter vector.

5 Laskin et al. (2022): Algorithm Distillation

Primary source. https://arxiv.org/abs/2210.14215

Mechanism and scope. Algorithm Distillation trains sequence models on learning histories so that adaptation can occur through context at inference. A history containing rewards and behaviour across episodes differs from a single successful trajectory.

Read, derive, investigate. Distinguish test-time context changes from gradient updates. Which data order exposes improvement rather than merely good behaviour? Evaluate on tasks held out from training and include a no-history ablation. A short sequence-classification toy is not proof that an optimiser has been fully recovered.

Connect the reading to your workbook

Run novelty, external return and coverage diagnostics together. For meta-learning, keep adaptation and evaluation samples separate, then compare full and first-order meta-gradients on a scalar problem before scaling up. Novelty, task learning and across-task adaptation answer different questions.

Evidence boundary. Source identities and the central claims above were checked against primary material. These annotations do not reproduce every proof, training run or empirical result. Use the original paper for theorem assumptions and experimental details; use `project/` and the workbook for the exact course implementation.

Statistics reading bridge

Adaptation steps are not independent new tasks. Read sample-mean uncertainty with tasks as units. Distinguish within-task adaptation randomness from across-task generalization; use exploration/meta-RL primary sources for the algorithm and the information it assumes.

Checked textbook locations: Siegrist § § 6.2, 8.1, PDF pp.817, 960. https://stats.libretexts.org/@go/page/10179. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.